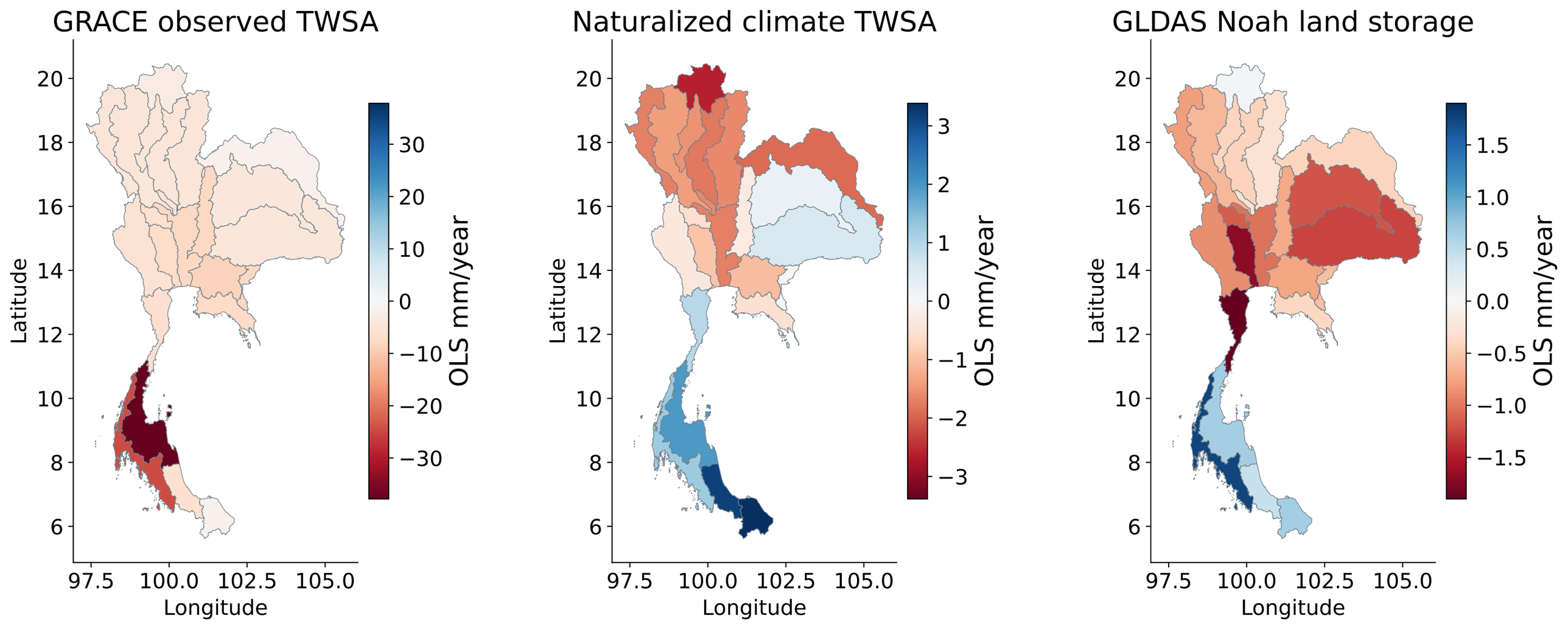
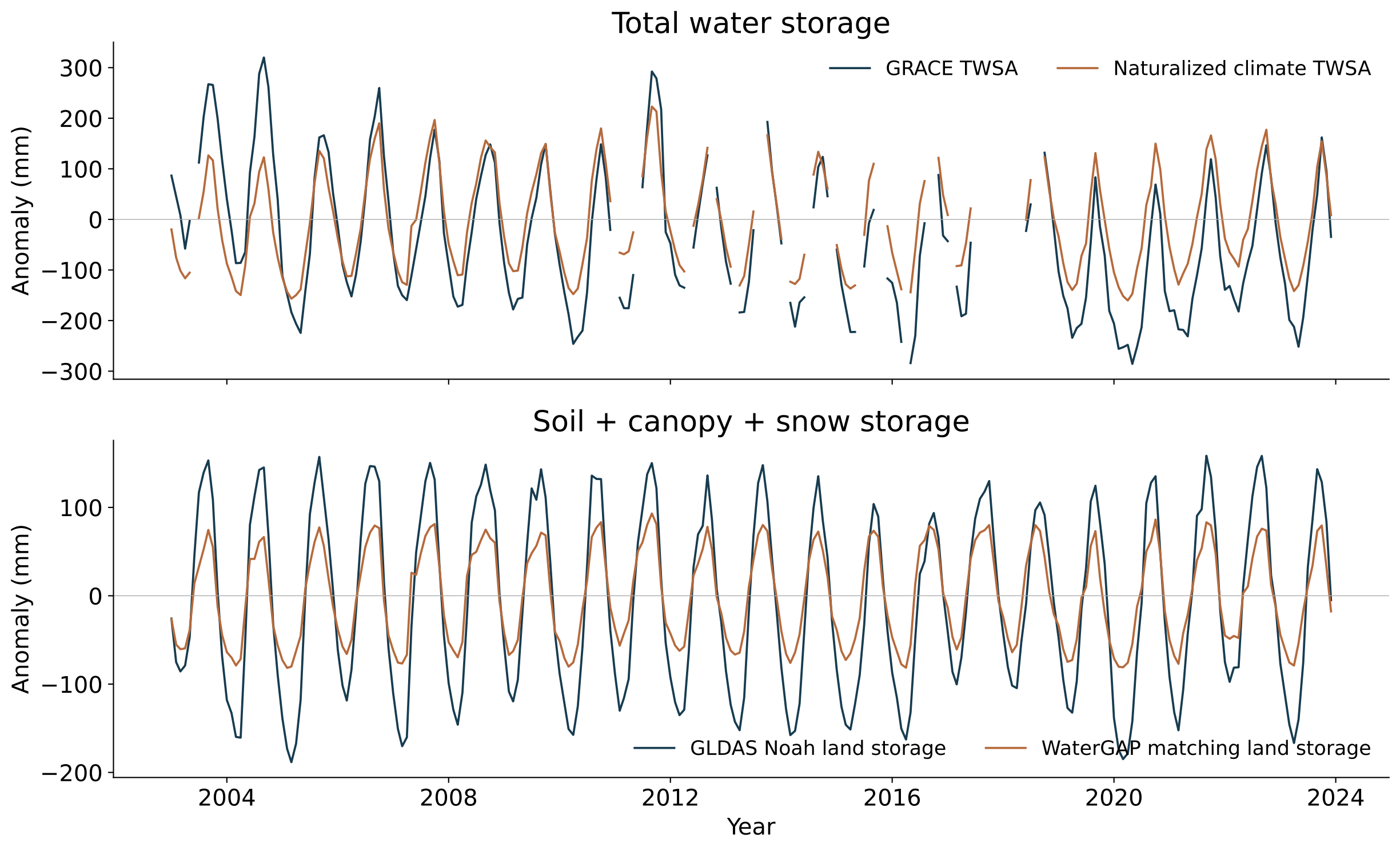

GRACE observed TWSA
Naturalized climate TWSA
GLDAS Noah land storage
Latitude
Longitude
OLS mm/year
20
18
16
14
12
10
8
6
97.5
100.0
102.5
105.0
30
20
10
0
−10
−20
−30
3
2
1
0
−1
−2
−3
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5


Total water storage
GRACE TWSA
Naturalized climate TWSA
Anomaly (mm)
300
200
100
0
−100
−200
−300
Soil + canopy + snow storage
GLDAS Noah land storage
WaterGAP matching land storage
Anomaly (mm)
100
0
−100
−200
2004
2008
2012
2016
2020
2024
Year

Naturalized parent Climate emulator

01 02 03 04
05 06 07 08
09 10 11 12
13 14 15 16
17 18 19 20
21 22

mm water equivalent

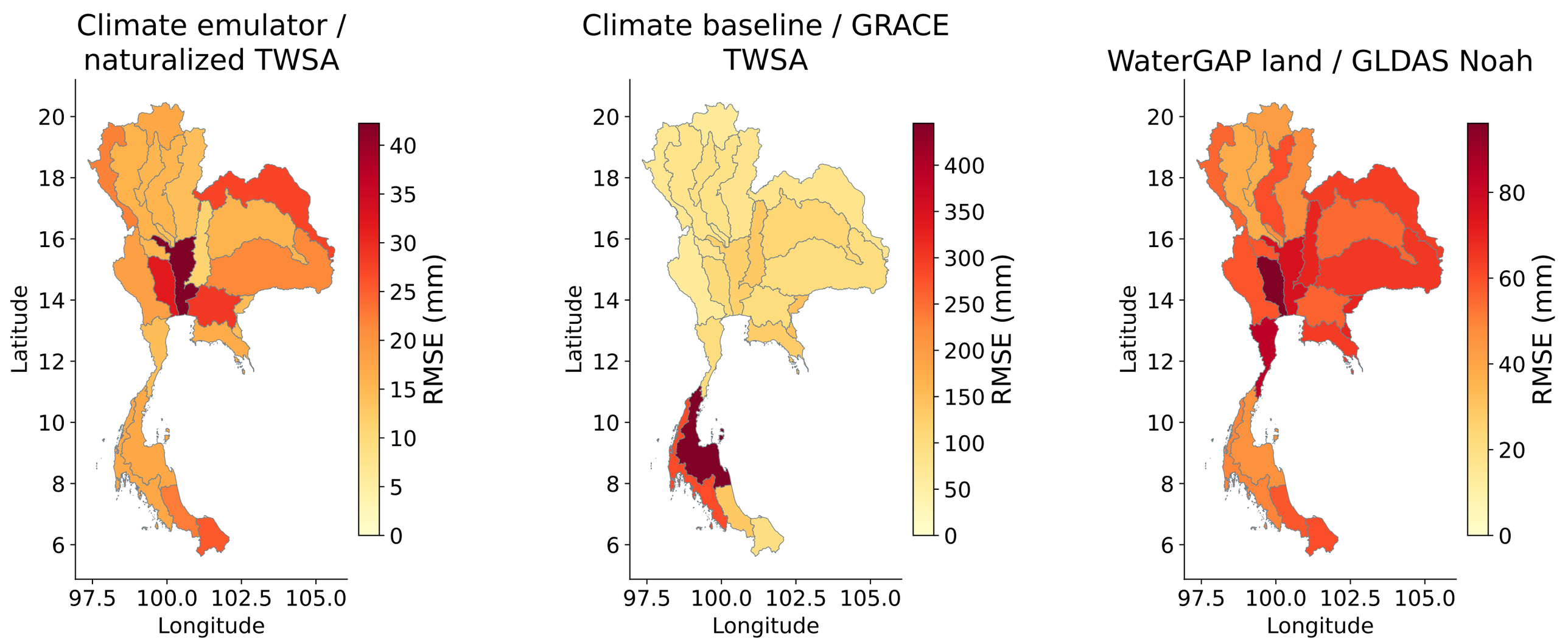
Calendar year · basin codes correspond to Appendix A

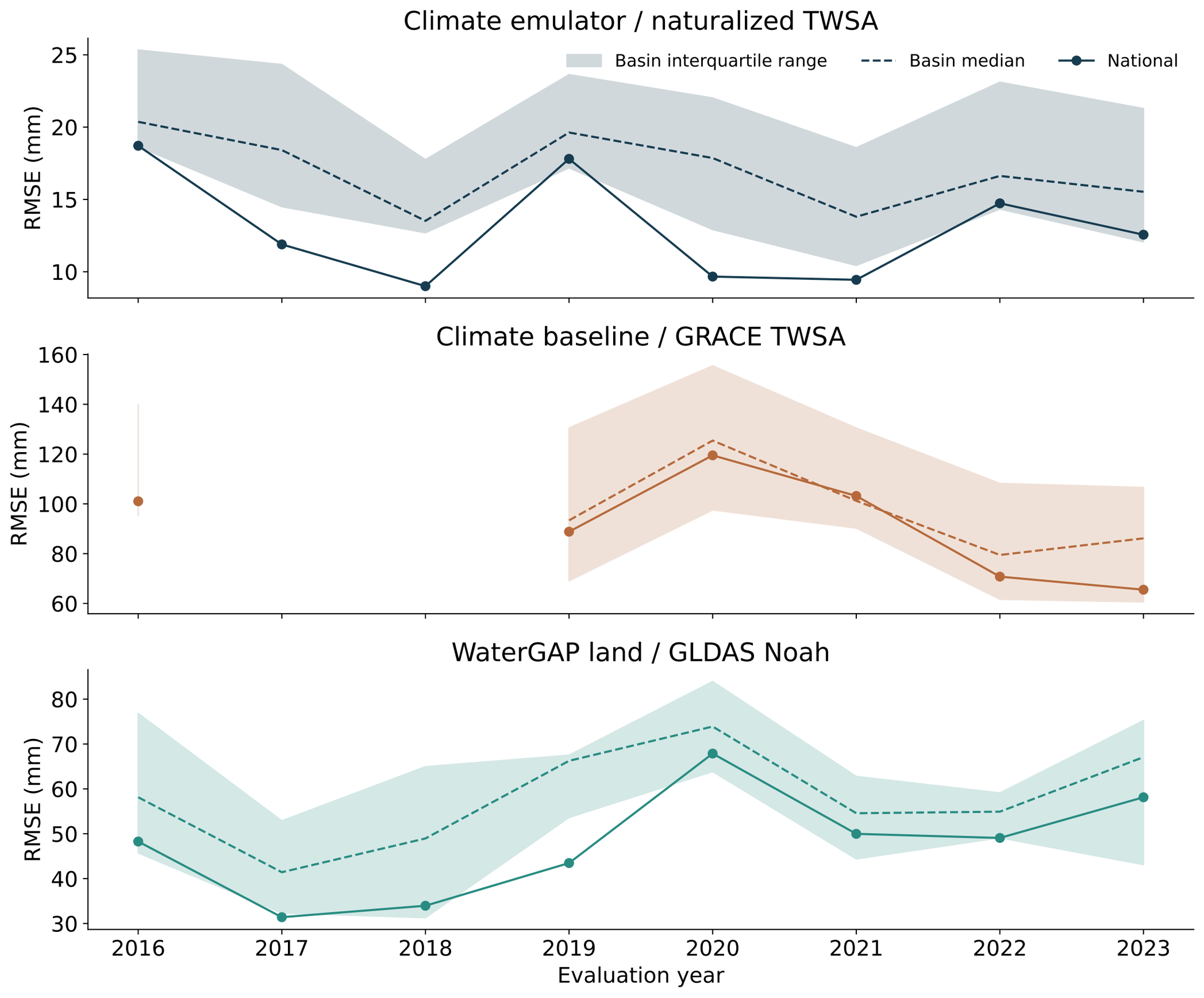
GLDAS land storage
Naturalized land storage

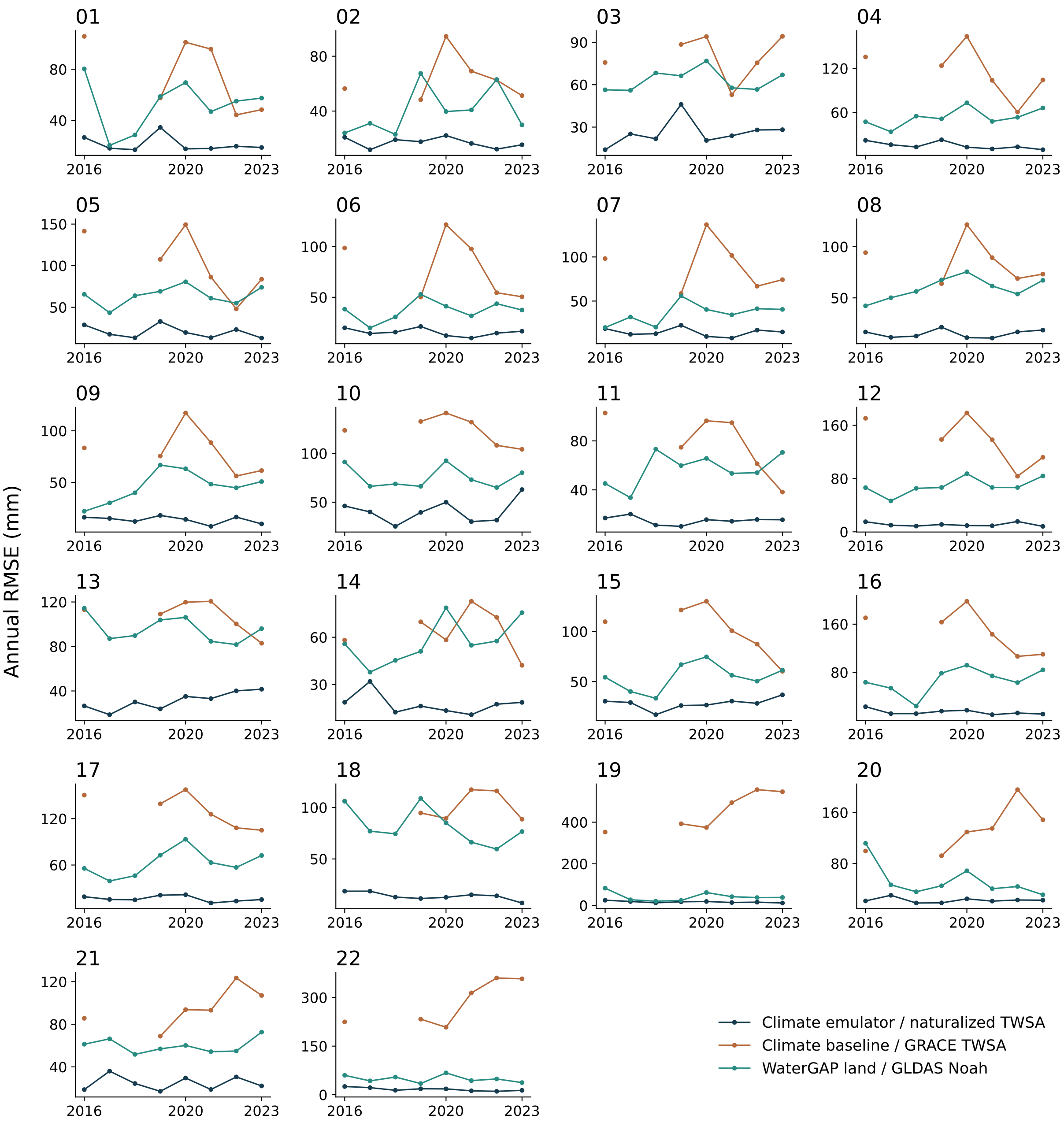
01 02 03 04
05 06 07 08
09 10 11 12
13 14 15 16
17 18 19 20
21 22

mm water equivalent

Calendar year · basin codes correspond to Appendix A


Climate emulator / naturalized TWSA
Climate baseline / GRACE TWSA
WaterGAP land / GLDAS Noah
Latitude
Longitude
RMSE (mm)
20
18
16
14
12
10
8
6
97.5
100.0
102.5
105.0
0
5
10
15
20
25
30
35
40
50
100
150
200
250
300
350
400
20
40
60
80


Climate emulator / naturalized TWSA
Basin interquartile range
Basin median
National
RMSE (mm)
25
20
15
10
Climate baseline / GRACE TWSA
RMSE (mm)
160
140
120
100
80
60
WaterGAP land / GLDAS Noah
RMSE (mm)
80
70
60
50
40
30
2016
2017
2018
2019
2020
2021
2022
2023
Evaluation year


01
02
03
04
05
06
07
08
09
10
11
12
13
14
15
16
17
18
19
20
21
22
Annual RMSE (mm)
2016
2020
2023
Climate emulator / naturalized TWSA
Climate baseline / GRACE TWSA
WaterGAP land / GLDAS Noah
Year · basin codes match the supplied CSV names